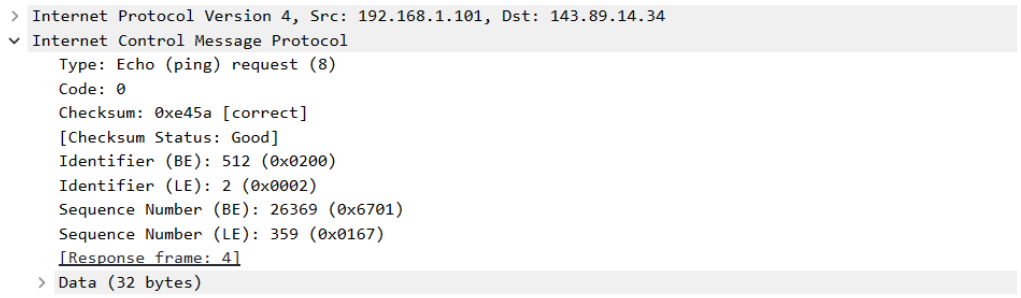
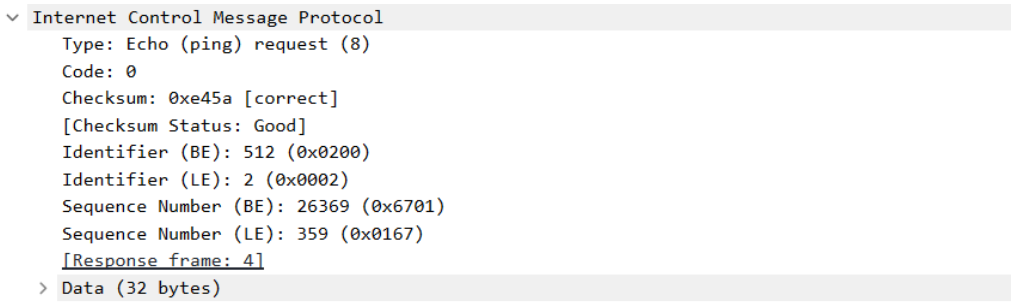
Assignment #8 - ICMP Lab

- **Question #1: What is the IP address of your host? What is the IP address of the destination host?**
  - Host IP Address: 192.168.1.101
  - Destination IP Address: 143.89.14.34



- **Question #2: Why is it that an ICMP packet does not have source & destination port numbers?**
  - ICMP packets do not have source & destination port numbers because it doesn't use the transport layer (TCP or UDP). ICMP is a network layer protocol and only uses IP addresses.
- **Question #3: Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number, and identifier fields?**
  - ICMP Type: Echo (ping) request (8)
  - ICMP Code Number: 0
  - Fields: Checksum, identifier (BE), identifier (LE), sequence number (BE), sequence number (LE)
    - BE → Big Endian
    - LE → Little Endian

        - LE is usually NOT actually part of the packet, but Wireshark shows it anyway.
            - The directions are unclear as whether or not to include Wireshark provided fields or not
    - Checksum Byte #: 2 bytes
    - Sequence Byte #: 2 bytes
    - Identifier Fields Byte #: 2 bytes



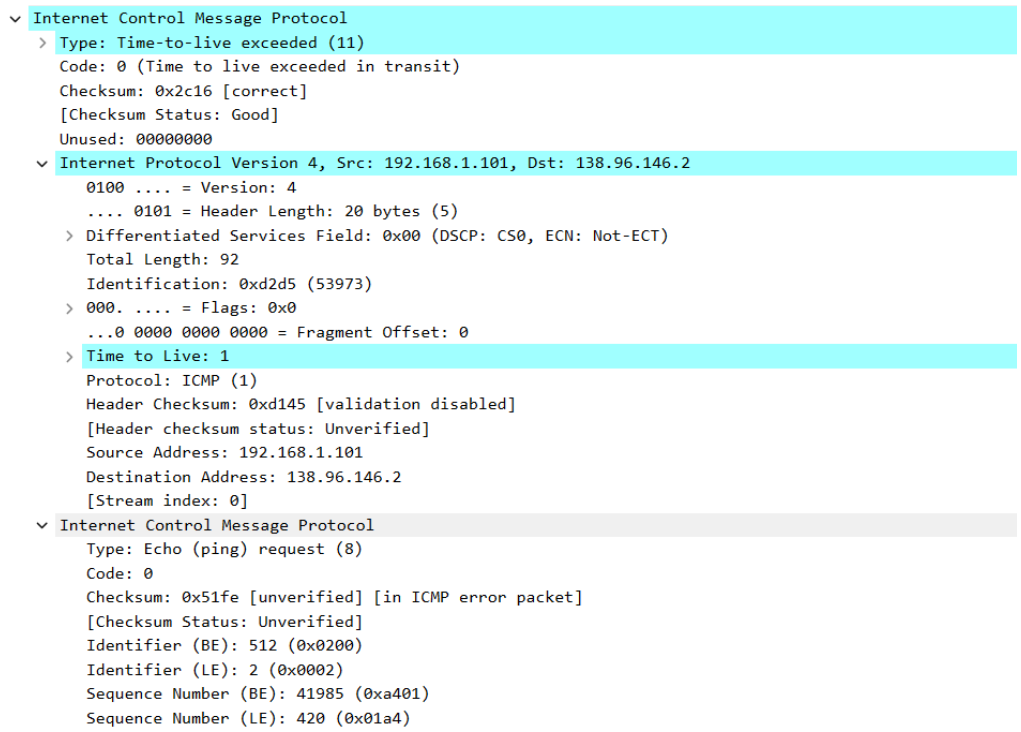
- **Question #4: Examine the corresponding ping reply packet. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number, and identifier fields?**
    - ICMP Type: Echo (ping) reply (0)
    - ICMP Code Number: 0
    - Fields: Checksum, identifier (BE), identifier (LE), sequence number (BE), sequence number (LE), Data
    - Checksum Byte #: 2
    - Sequence Number Byte #: 2
    - Identifier Field Byte #: 2

```
Internet Control Message Protocol
    Type: Echo (ping) reply (0)
    Code: 0
    Checksum: 0xec5a [correct]
    [Checksum Status: Good]
    Identifier (BE): 512 (0x0200)
    Identifier (LE): 2 (0x0002)
    Sequence Number (BE): 26369 (0x6701)
    Sequence Number (LE): 359 (0x0167)
    [Request frame: 3]
    [Response time: 413.442 ms]
  > Data (32 bytes)
```

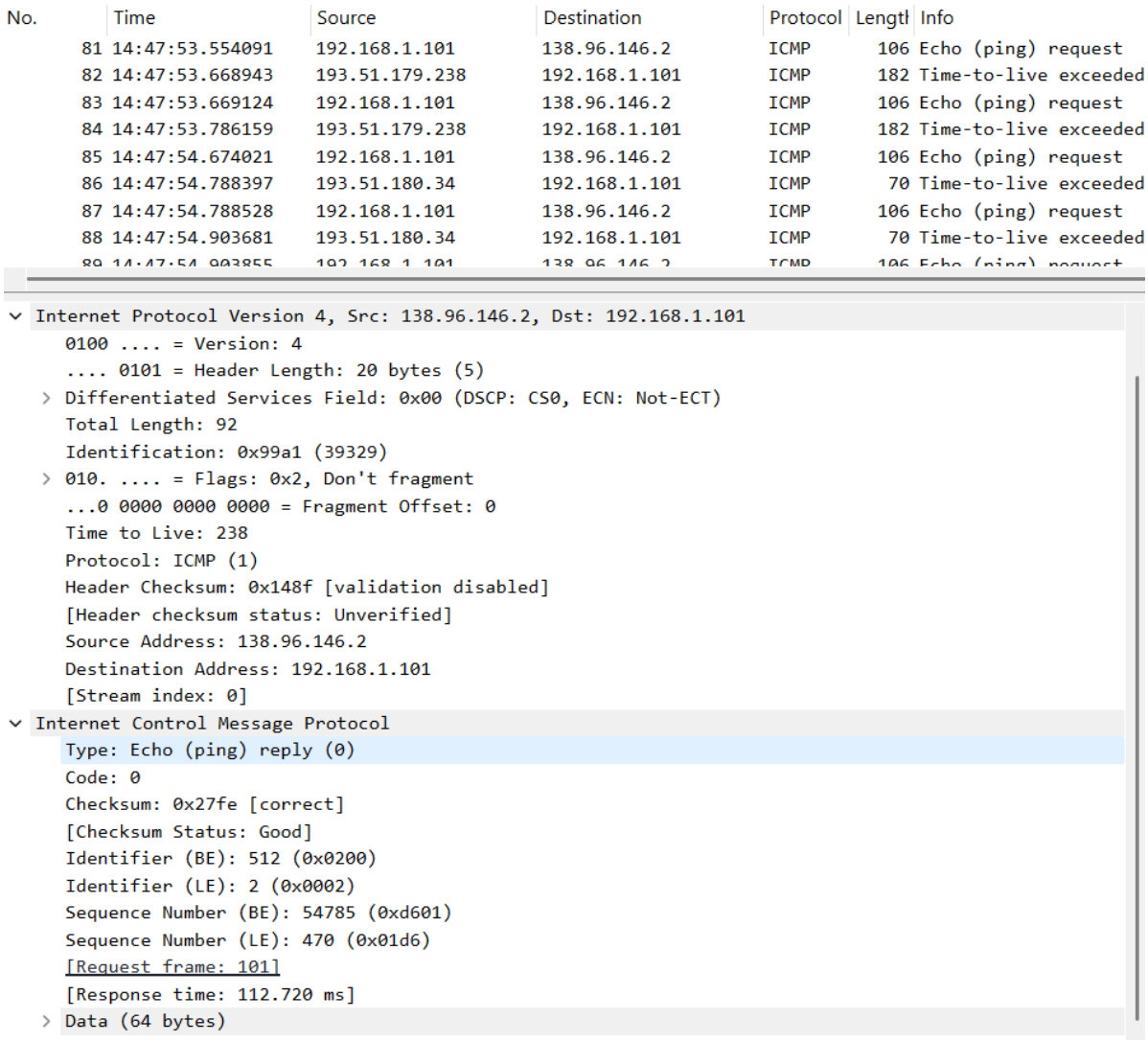
- **Question #5: What is the IP address of your host? What is the IP address of the target destination host?**
  - Host IP Address: 192.168.1.101
  - Destination IP Address: 138.96.146.2

```
Internet Protocol Version 4, Src: 138.96.146.2, Dst: 192.168.1.101
```

- **Question #6: If ICMP sent UDP packets instead (as in Unix/Linux), would the IP protocol number still be 01 for the probe packets? If not, what would it be?**
  - If ICMP sent UDP packets instead, the IP protocol number would be 17 instead of 01.
- **Question #7: Examine the ICMP echo packet in your screenshot. Is this different from the ICMP ping query packets in the first half of this lab? If yes, how so?**
  - The ICMP echo packets are the same.
- **Question #8: Examine the ICMP error packet in your screenshot. It has more fields than the ICMP echo packet. What is included in those fields?**
  - Differences:
    - Type: Time to live exceeded (11)
    - Unused: 00000000
  - It also includes the IP header and the first 8 bytes of the original ICMP packet that received an error



- **Question #9: Examine the last three ICMP packets received by the source host. How are these packets different from the ICMP error packets? Why are they different?**
    - The last three ICMP packets that the source host received have message type 0 (echo reply) rather than time-to-live exceeded (11).
    - They are different because the ICMP packets it all the way to the destination host without having the TTL expire.



No.	Time	Source	Destination	Protocol	Length	Info
81	14:47:53.554091	192.168.1.101	138.96.146.2	ICMP	106	Echo (ping) request
82	14:47:53.668943	193.51.179.238	192.168.1.101	ICMP	182	Time-to-live exceeded
83	14:47:53.669124	192.168.1.101	138.96.146.2	ICMP	106	Echo (ping) request
84	14:47:53.786159	193.51.179.238	192.168.1.101	ICMP	182	Time-to-live exceeded
85	14:47:54.674021	192.168.1.101	138.96.146.2	ICMP	106	Echo (ping) request
86	14:47:54.788397	193.51.180.34	192.168.1.101	ICMP	70	Time-to-live exceeded
87	14:47:54.788528	192.168.1.101	138.96.146.2	ICMP	106	Echo (ping) request
88	14:47:54.903681	193.51.180.34	192.168.1.101	ICMP	70	Time-to-live exceeded

```
Internet Protocol Version 4, Src: 138.96.146.2, Dst: 192.168.1.101
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 92
    Identification: 0x99a1 (39329)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 238
    Protocol: ICMP (1)
    Header Checksum: 0x148f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 138.96.146.2
    Destination Address: 192.168.1.101
    [Stream index: 0]
Internet Control Message Protocol
    Type: Echo (ping) reply (0)
    Code: 0
    Checksum: 0x27fe [correct]
    [Checksum Status: Good]
    Identifier (BE): 512 (0x0200)
    Identifier (LE): 2 (0x0002)
    Sequence Number (BE): 54785 (0xd601)
    Sequence Number (LE): 470 (0x01d6)
    [Request frame: 101]
    [Response time: 112.720 ms]
  > Data (64 bytes)
```

- **Question #10: Within the tracert measurements, is there a link whose delay is significantly longer than others? Refer to the screenshot in Figure 4, is there a link whose delay is significantly longer than others? On the basis of the router names, can you guess the location of the two routers on the end of this link?**
    - There is a link whose delay is significantly longer, and all of the links after this one also maintain a significantly longer delay the previous ones.
    - I'd guess that these routers are in Europe instead of America, thus causing the delay

```
 1    13 ms    12 ms    13 ms  10.216.228.1
 2    21 ms    14 ms    13 ms  24.218.0.153
 3    12 ms    11 ms    13 ms  bar01-p4-0.wsfdhe1.ma.attbb.net [24.128.190.197]
 4    16 ms    16 ms    15 ms  bar02-p6-0.ndhmhe1.ma.attbb.net [24.128.0.101]
 5    15 ms    15 ms    15 ms  12.125.47.49
 6    17 ms    17 ms    17 ms  12.123.40.218
 7    22 ms    23 ms    22 ms  tbr2-cl1.n54ny.ip.att.net [12.122.10.22]
 8    23 ms    23 ms    23 ms  ggr2-p3120.n54ny.ip.att.net [12.123.3.109]
 9    26 ms    21 ms    25 ms  att-gw.nyc.opentransit.net [192.205.32.138]
10    98 ms    98 ms    96 ms  P4-0.PASCR1.Pastourelle.opentransit.net [193.251.241.133]
11    97 ms    98 ms    98 ms  P9-0.AUVCR1.Aubervilliers.opentransit.net [193.251.243.29]
12    98 ms    98 ms   108 ms  P6-0.BAGCR1.Bagnolet.opentransit.net [193.251.241.93]
13   104 ms   106 ms   103 ms  193.51.185.30
14   114 ms   114 ms   117 ms  grenoble-pos1-0.cssi.renater.fr [193.51.179.238]
15   114 ms   115 ms   114 ms  nice-pos2-0.cssi.renater.fr [193.51.180.34]
16   129 ms   114 ms   118 ms  inria-nice.cssi.renater.fr [193.51.181.137]
17   113 ms   114 ms   112 ms  www.inria.fr [138.96.146.2]
```